QUANTIFYING THE SPATIAL LANDSCAPE OF NON-MUSCLE INVASIVE BLADDER CANCER TUMOR IMMUNE MICROENVIRONMENT

TECHNICAL REPORT

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ABSTRACT

This report explores the application of convolutional neural networks (CNNs) by transfer learning on Imaging Mass Cytometry (IMC) data to better understand the tumor microenvironment of bladder cancer (TME), its response to immunotherapy and the biological determinants of treatment outcome. IMC provides spatially resolved, multi-marker single-cell data, enabling characterization of immune, epithelial, and stromal components. I benchmark multiple deep learning architectures to classify clinically relevant attributes (grade, response to therapy) from IMC regions of interest (ROIs). My findings show that both modified ResNet architectures and feature-extraction-based methods exhibit promising performance, with accuracy ranging from $\tilde{5}5$ to 65% depending on the task and sample type. Challenges remain due to limited sample size and high dimensionality, but the results indicate that deep learning can extract biologically meaningful features from IMC images. Future work will involve larger datasets (COSMX, MICS), HE-based pipelines, and testing on public datasets.

Keywords First keyword · Second keyword · More

1 Introduction

Immunotherapy has been shown to extend survival in multiple cancers, including bladder cancer (BC)Waldman et al. [2020]. It acts by adjusting the patient's immune system to trigger an anti-tumor reaction. The effectiveness of immune cells, determined by their effector activity and the presence of inhibitory receptors, substantially influences their capacity to initiate and sustain immune responsesWherry and Kurachi [2015]. Functional effector mechanisms can be mediated both directly and indirectly through either cell-cell interactions or cytokine signaling. Hence, to gain a deeper understanding of the importance of the functional status of immune cells for treatment outcomes, it is imperative to perform detailed quantitative spatial analyses of the tumor microenvironment (TME). Transfer learning offers a powerful approach for conducting such spatial analyses by leveraging knowledge from large, pretrained models and adapting it to complex histological dataBozinovski [2020]. This strategy not only enables accurate classification of tissue regions according to their immunophenotypes but also has the potential to uncover deeper biological patterns underlying immune function within the TME.

This study investigates whether deep learning can characterize the bladder cancer TME using IMC data and whether CNN-derived spatial features can help elucidate biological patterns related to tumor grade, treatment response, and immunological states relevant to BCG therapy. Using a 39-channel IMC dataset I evaluated several neural network architectures—including channel-adapted ResNet variants and marker-grouped models—and benchmark their ability to classify clinically relevant targets. My goal is to assess the feasibility of deep learning for spatial immunophenotyping in NMIBC and to provide a methodological foundation for future prediction of BCG responsiveness using IMC-derived spatial signatures. This project is distinct from other deep learning for visual recognition projects in that the degree to which samples can be realistically classified is not guaranteed going into the project. Therefore, the success of this project and the models being tested is not determined solely by the accuracy of the classifiers being tested, but also by the degree to which they can be used to interogate the underlying biology that is distinct between samples.

2 Related Work

2.1 Non-Muscle Invasive Bladder Cancer

Non–muscle-invasive bladder cancer (NMIBC) represents the majority of newly diagnosed bladder cancer cases and is characterized by a high rate of recurrence and progression despite treatment. Standard-of-care treatment consists of transurethral resection of the tumor followed by intravesical instillation of Bacillus Calmette–Guérin (BCG), an immunotherapy that triggers a broad activation of innate and adaptive immune responses. However, clinical outcomes remain heterogeneous: up to 40% of patients experience high-grade recurrence after BCG therapy, and a significant subset eventually progresses to muscle-invasive diseasePrip et al. [2025]. Understanding the determinants of BCG responsiveness has therefore emerged as a major clinical priority.

2.2 Tumor Microenvironment

Recent transcriptomic and multi-omics studies have revealed that BCG failure is tightly linked to the functional state of immune cells within the tumor microenvironment (TME)Strandgaard et al. [2022]. Elevated CD8 positive T-cell exhaustion after BCG is strongly associated with high-grade recurrence and poor recurrence-free survival, and pretreatment tumors enriched for cell-cycle and immunoactive gene programs tend to develop exhausted phenotypes following therapy . These findings have highlighted the need to understand the response of BCG not only through genomics or bulk expression, but through spatially resolved analysis of immune–tumor interactions at single-cell resolution.

Imaging Mass Cytometry (IMC) enables multiplexed detection of dozens of proteins in tissue sections while preserving spatial architecture, providing an unprecedented view of the TMEAli et al. [2020]. IMC studies in lung adenocarcinoma have shown that spatial arrangements of immune and stromal populations—rather than cell frequencies alone—strongly predict clinical outcomtHIe, and deep learning applied to raw IMC images can identify patients likely to progress after surgery with >90% accuracySorin et al. [2023]. This underscores the power of spatial omics combined with CNNs for predicting clinically meaningful phenotypes .

2.3 Convolutional Neural Networks and Pytorch

Convolutional neural networks (CNNs) have emerged as a cornerstone of modern image analysis, offering a powerful framework for detecting hierarchical patterns in high-dimensional biomedical data. Their architecture; built around convolutional filters, nonlinear transformations, and progressively abstract feature representations; makes them uniquely suited for learning spatially organized signals such as cellular niches, gradients of protein expression, and tissue-level structural motifsSorin et al. [2023]. In the context of multiplexed imaging platforms like IMC, CNNs can efficiently parse complex spatial relationships across dozens of protein channels without requiring manual feature engineering, enabling unbiased discovery of spatially encoded biological signatures.

Pytorch provides a flexible and computationally efficient environment for implementing CNNs tailored to these dataPaszke et al. [2019]. Its modular design facilitates rapid experimentation with model architectures, optimization strategies, and training procedures, while its GPU-accelerated operations allow for scalable analysis of large, high-resolution imaging datasets. Together, CNNs and Pytorch form a robust methodological foundation for extracting spatial features from IMC images, supporting both supervised tasks such as immunophenotype classification and exploratory analyses aimed at uncovering latent biological structure within the TME.

3 Methods

3.1 Data

The dataset consists of Imaging Mass Cytometry (IMC) images acquired from bladder tumor specimens stained with a panel of 39 metal-conjugated antibodies. These markers span epithelial, immune, and stromal compartments, including cytokeratins, T-cell markers, macrophage-associated proteins, immune checkpoints, and proliferation markers. Each image corresponds to a 1 mm^2 circular region of interest (ROI) selected from the whole tissue section, with an average of three ROIs per tissue section for each patient.

Due to varying staining robustness and biological relevance, the full panel was filtered to a curated subset of 29 high-quality channels for downstream modeling. This selection was guided by separate single-cell proteomics work that identified individual cells, their lineages, and compartments (Fig. 1). Markers failing QC in this pipeline were excluded from the deep-learning analysisnoa

3.2 Image Preparation and Channel Processing

IMC images contain substantially more channels than conventional RGB images, which necessitates explicit strategies for adapting high-dimensional marker data to convolutional neural networks (CNNs). I therefore implemented three complementary ways of presenting the 29 IMC channels to a backbone classifier (Fig. 1).

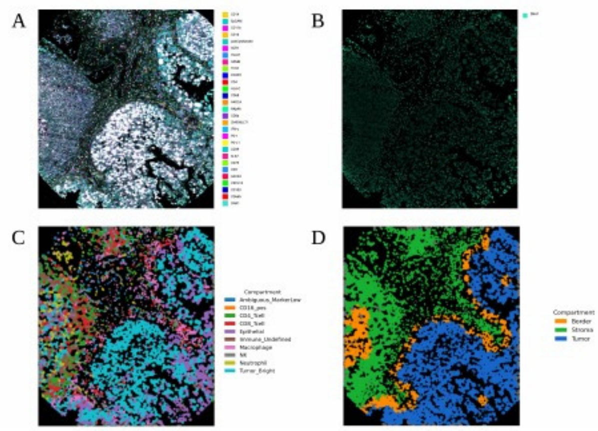

Figure 1: A. All marker channels that passed quality control, visualised on an example region of interest (ROI). B. DNA channel. C. Major cell lineages assigned using a single-cell proteomics reference. D. Compartment annotations for the ROI.

Direct channel compression. The full 29-channel IMC tensor is mapped to a 3-channel pseudo-RGB representation using a learned 1×1 convolutional *ChannelAdapter*, which linearly mixes markers at each pixel into three channels compatible with standard ImageNet-pretrained CNNs.

Biologically grouped compression. Channels are grouped into epithelial, lymphoid, and myeloid compartments based on marker function. A biologically constrained adapter compresses each group into its own pseudo-channel, preventing mixing across compartments and yielding three interpretable channels (epithelial, lymphoid, myeloid).

Full-channel convolution. The first convolutional layer of the CNN backbone is modified to accept all 29 channels directly, avoiding any explicit compression and allowing the network to learn marker-wise and spatial interactions in a high-dimensional input space.

Across all approaches, images are per-channel z-score normalised, padded to a fixed canvas, and cropped into patches for training. Standard data augmentation (random flips and spatial cropping, with intensity normalisation) is applied to improve generalisation.

3.3 CNN Architectures

Each of the channel-processing strategies in Section 3.2 is paired with a corresponding neural network design, resulting in four architectures tailored to the IMC data (Fig. 2).

Architecture 1 — ChannelAdapter + ResNet18. Architecture 1 applies the direct channel compression strategy, feeding the 3-channel pseudo-RGB output of the ChannelAdapter into an ImageNet-pretrained ResNet18 backbone. Training proceeds in a transfer-learning regime with the classification head and adapter trained first, followed by progressive unfreezing of deeper residual blocks.

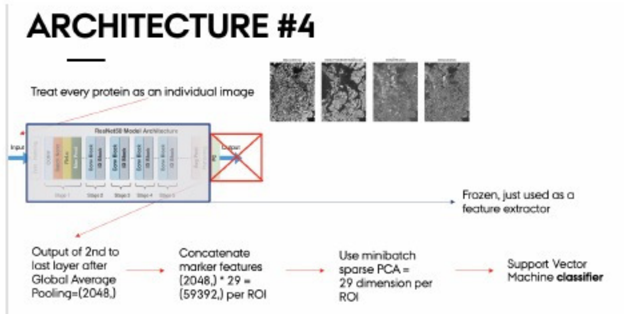

Figure 2: Schematic overview of Architecture 4: marker-wise feature extraction with a frozen ResNet backbone, sparse PCA, and SVM classifier.

Architecture 2 — Biologically grouped channels + ResNet18. Architecture 2 uses the biologically grouped compression pathway, in which epithelial, lymphoid, and myeloid channels are compressed into three interpretable pseudo-channels. These are then passed to an ImageNet-pretrained ResNet18, again with staged unfreezing. This design preserves the biological compartment structure while leveraging the same backbone and optimisation scheme as Architecture 1.

Architecture 3 — 29-channel ResNet. Architecture 3 implements the full-channel convolution strategy by modifying the first layer of ResNet18 to accept the 29-channel IMC input directly. The remaining backbone and training schedule are analogous to Architectures 1 and 2, but without any intermediate channel compression.

Architecture 4 — Marker-wise feature extraction + SPCA + SVM. In Architecture 4, each IMC marker is treated as an independent grayscale image. For each channel, the image is replicated to three channels and passed through a frozen ResNet backbone to extract a 2048-dimensional feature vector via global average pooling. The 29 channel-wise feature vectors are concatenated into a single 59,392-dimensional representation per ROI. MiniBatch sparse PCA (SPCA) is fitted on the training set to obtain a low-dimensional latent space, and a radial basis function support vector machine (RBF-SVM), combined with random over-sampling to address class imbalance, is trained on these SPCA features. This architecture decouples feature extraction from classification and provides a modular, interpretable alternative to end-to-end CNN training.

All architectures are implemented in PyTorch, trained with cross-entropy loss for binary classification, and optimised using AdamW with early stopping based on validation AUROC. Patient-level stratified 5-fold cross-validation is used in all experiments to avoid information leakage between tiles from the same individual.

3.4 Prediction Tasks

The primary supervised task explored in this study was tumor grade classification, using histopathological labels corresponding to pre- and post-BCG samples. Additional exploratory tasks included prediction of 2-year treatment response and distinction between pre-BCG versus post-BCG tissue states. These tasks represent clinically meaningful phenotypes that may be reflected in spatial immune architecture.

3.5 Evaluation

Performance was assessed using fivefold cross-validation at the patient level to avoid data leakage across ROIs from the same tumor. Metrics included accuracy, precision, recall, and mean performance across folds. For Architecture 4, principal component embeddings were visualized using UMAP to assess feature separation between classes. Confusion matrices and performance distributions across folds were examined to evaluate model stability and robustness.

4 Results

4.1 IMC Dataset Quality and Structure

IMC images exhibited high-quality spatial organization with distinct expression patterns across epithelial, immune, and stromal markers. Preprocessing successfully harmonized intensity distributions across regions of interest (ROIs), and

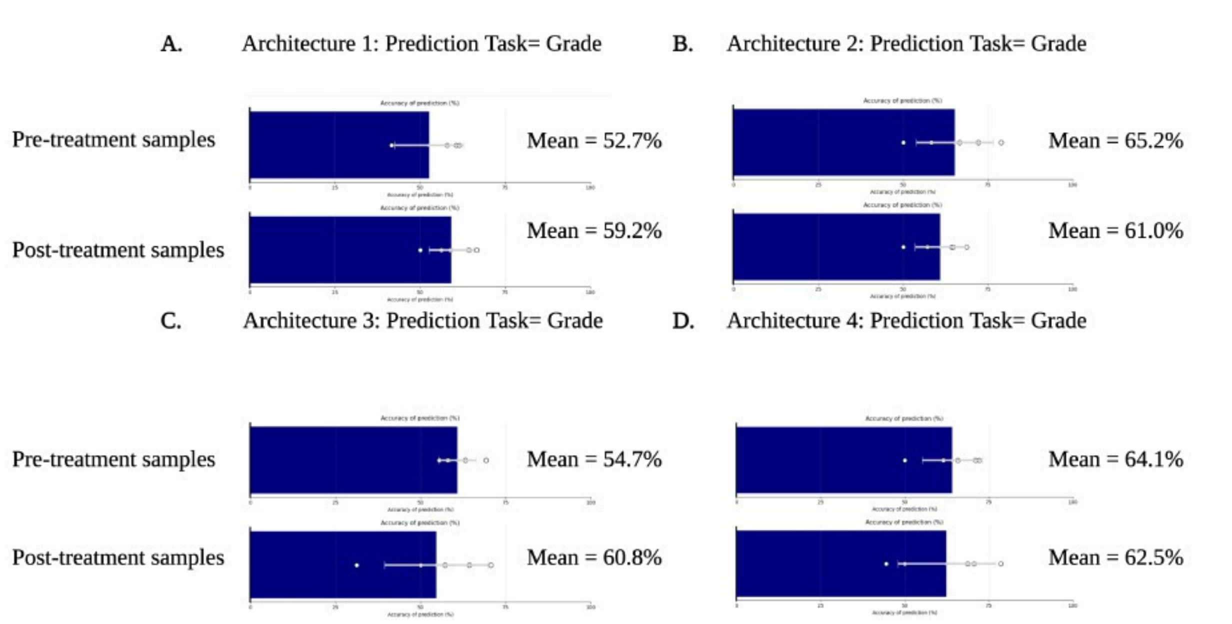

Figure 3: Mean accuracies for grade prediction across architectures across folds.
A. Architecture 1 — ChannelAdapter + ResNet18.
B. Architecture 2 — Biologically grouped channels.
C. Architecture 3 — Modified first convolution.
D. Architecture 4 — Marker-wise feature extraction.

marker–marker correlations reflected expected biological relationships, such as strong co-expression among epithelial markers and inverse relationships between immune and tumor compartments (Fig. 1).

4.2 Classification Performance

Classification performance varied across architectures and preprocessing strategies.

The simple channel adapter coupled with progressively unfrozen ResNet18 had the worst performance (Architecture 1) ((Fig. 3 A). The Biologically Grouped model (Architecture 2) and the modified-channel models (Architecture 3) achieved mean accuracies of approximately 65.2% and 61% respectively for pre-treatment grade prediction and 60.8% and 58% respectively for post-treatment grade, slightly exceeding random baseline performance (Fig. 3 B) and (Fig. 3) C). Architecture 4 achieved slightly higher and more even accuracy across the timepoints, reaching roughly 64% for pre-BCG samples and 62% for post-BCG. Architecture 4 was thus chosen to proceed to attempt to classify ROIs for 2 year reponse to treatment.

Prediction of 2-year treatment response proved similarly challenging to grade. This metric is a binary metric that communicates whether or not the patient experiences a high grade disease reoccurrence or progression within 2 years of BCG treatment. Pre-treatment response prediction had a mean accuracy of 45.1%, indicating worse than random performance, and post-treatment response prediction had a mean accuracy of 63%.(Fig. 4 A). This is the first time there is a meaningful distinction between the pre and post treatment samples. This indicates that while accuracy may be low and vary between folds greatly, and area under aggregate receiver operator curve is worse than random, there is potentially some meaningful treatment remodeling that is being detected. This motivated using timing of samples as a classifier. Timing prediction had a mean accuracy of 59.9%, with heterogenouse per-fold AUROC, but an aggregate AUROC of 0.592.(Fig. 4 C and D)

4.3 Biological Interpretation of Learned Features

To explore how individual markers contributed to the Architecture 4 classifier, UMAP was applied to the 2048-dimensional feature vectors extracted from the frozen ResNet18 backbone for each IMC channel (Fig. 5). Markers

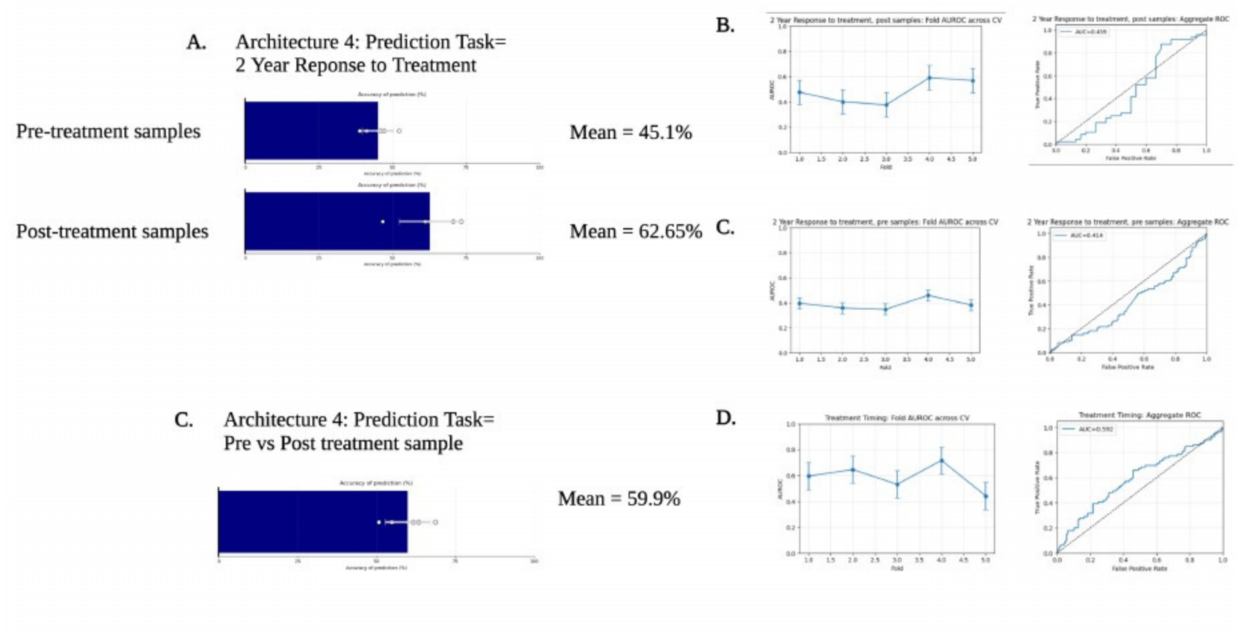

Figure 4: A. Performance of Architecture 4 for predicting 2-year response to treatment. B. Per-fold Area Under the Receiver Operating Characteristic Curve (AUROC) and aggregate ROC for pre-treatment samples. C. Per-fold AUROC and aggregate ROC for post-treatment samples.

associated with immune remodeling, like CD8A (cytotoxic T cells) and CD68 (macrophage lineage), showed coherent structure in UMAP space, indicating that the model captures biologically meaningful variation in these cell populations. Epithelial and differentiation-associated markers (e.g. PANCYTOKERATIN, GATA3) also formed smooth gradients consistent with tissue composition.

However, despite clear organization within each marker's feature manifold, there was no observable separation between pre- and post-treatment samples. This suggests that while the marker-wise embeddings reflect underlying tissue biology and immune architecture, these features alone do not encode systematic shifts attributable to treatment timing in this cohort.

5 Discussion

The primary aim of this study was to evaluate whether convolutional neural networks can learn biologically meaningful spatial representations from IMC data and whether these representations can support prediction of clinically relevant phenotypes in NMIBC. Although overall classification performance remained modest, several important insights emerged regarding both the feasibility and the limitations of deep learning on small, high-dimensional spatial omics datasets.

Challenges in Label Alignment and Tumor Heterogeneity

Tumor grade was initially expected to be the most tractable prediction target; however, performance across all architectures remained close to random. This outcome is biologically and methodologically consistent with the nature of the IMC acquisition strategy. IMC ROIs sample only small subregions of the tumor, whereas pathological grade is assigned at the whole-slide level and reflects the highest-grade region present anywhere in the specimen. Given the well-documented spatial heterogeneity of bladder tumors, many IMC ROIs likely do not contain the features that informed the original grade annotation. As such, poor performance likely reflects a mismatch between the spatial field of view and the ground-truth label rather than model failure. Future work using whole-slide H&E images—where spatial coverage matches the annotated phenotype—or using ROI-level expert grading would provide a more faithful benchmark.

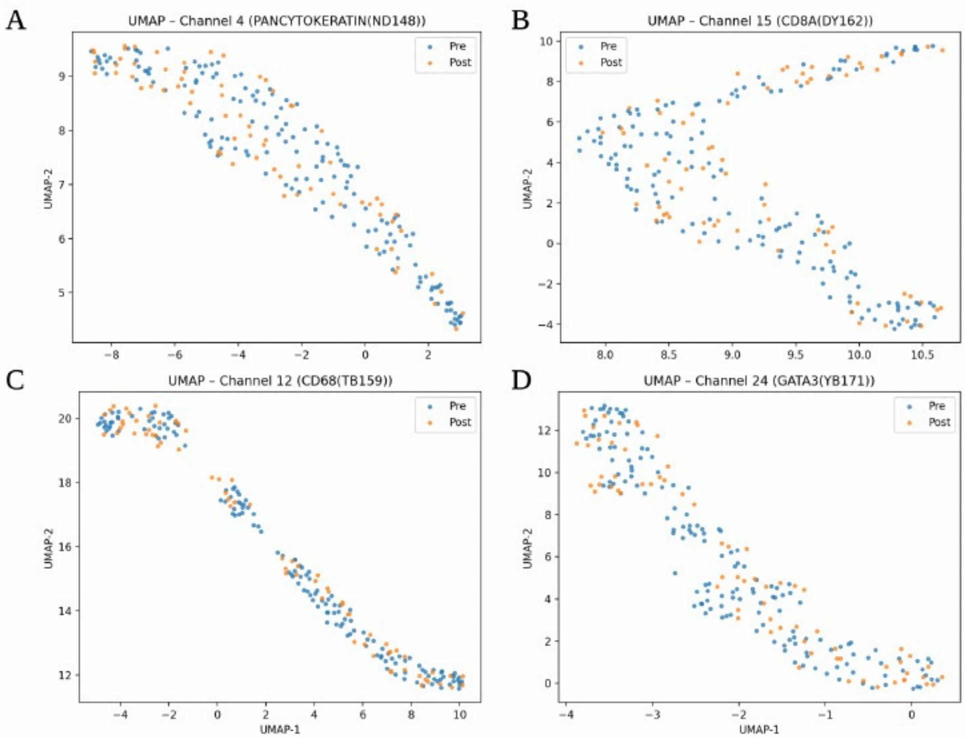

Figure 5: UMAP projections of marker-wise ResNet18 feature embeddings (Architecture 4) for Fold 0, illustrating representative channels: **A** PANCYTOKERATIN (epithelial), **B** CD8A (cytotoxic T cells), **C** CD68 (macrophages), and **D** GATA3 (luminal/Th2-associated). Points are coloured by treatment timing (Pre vs. Post).

Architecture Performance and Representational Insights

Among the models tested, the marker-wise feature extraction approach (Architecture 4) produced the most stable performance across tasks. While not dramatically outperforming the end-to-end CNNs, it offered two advantages: (i) robustness across folds despite the small dataset, and (ii) interpretability through channel-specific embeddings. The ability to inspect marker-wise UMAP projections revealed coherent biological structure, including smooth gradients for epithelial markers and clustered variation in CD8A and CD68 features, indicating that the pretrained backbone successfully captures lineage-associated spatial patterns.

Importantly, these embeddings did *not* separate pre- and post-treatment samples, suggesting that BCG-induced remodeling—while biologically meaningful—is subtle relative to inter-patient variability and not linearly encoded in marker-wise spatial features alone. This reinforces the need for larger sample sizes and potentially multichannel or cell-interaction–aware models.

Predicting Treatment Response and Timing

An unexpected but informative observation was that post-treatment samples yielded the highest performance for predicting 2-year response. Although accuracy remained modest, this suggests that BCG exposure amplifies or unmasks spatial immune patterns associated with treatment outcome. This aligns with prior evidence that BCG response is linked to the emergence of exhausted CD8 T-cell states, altered macrophage polarization, and shifts in epithelial–immune spatial Wang et al. [2021] Strandgaard et al. [2022]. While the models here could not reliably predict treatment response, they appear sensitive to early signals of post-therapy immune remodeling.

The ability to classify treatment timing with accuracy approaching 60% further supports the idea that BCG induces reproducible, if subtle, spatial changes in the TME. Although far from clinically actionable, these results demonstrate that CNN-based feature extraction can detect shifts in tissue architecture associated with therapy, even when prediction of clinical endpoints remains challenging.

Limitations and Future Directions

This study is constrained primarily by dataset size, label quality, and the mismatch between IMC's microscopic scope and whole-tumor clinical labels. Several directions may substantially improve model performance and interpretability:

- **Larger datasets and pooled cohorts:** Incorporating COSMX, MICS, or external IMC datasets would enable training of deeper models and address overfitting.
- **Cell-level and graph-based modeling:** Spatial graphs, cell–cell interaction features, and graph neural networks may capture microarchitectural remodeling more effectively than pixel-based CNNs.
- **Multimodal integration:** Joint modeling of IMC with H&E whole-slide imaging or transcriptomic profiling may better anchor spatial features to clinical phenotypes.
- **ROI selection strategies:** Using multiple ROIs per slide or active-learning strategies could mitigate the grade-label misalignment observed here.
- **Interpretability frameworks:** Channel attribution, saliency mapping, and marker-synergy analysis could help identify which biological programs drive the learned representations.

Conclusion

Despite limited predictive performance, the models developed here demonstrate that deep learning can extract biologically meaningful spatial features from IMC images and that these features reflect underlying immune and epithelial organization within the NMIBC TME. While classification of grade and treatment response remains challenging, early evidence suggests that therapy-induced remodeling leaves detectable signatures in spatial protein expression. With larger datasets, improved alignment of labels to ROIs, and integration of cell-interaction features, deep learning has strong potential to illuminate the spatial determinants of BCG responsiveness and advance precision immuno-oncology in bladder cancer.

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